What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Agents treating Helminthic and Protozoal infections

Last updated June 13, 2025 · 10 min read · CC BY-SA 4.0

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Adapted from: Drugs to Treat Protozoal and Helminthic Infections

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action of antihelminthic and antiprotozoal drugs.
2. List the indications for anthelminthic and antiprotozoal drugs.
3. Recognize th … d with anthelminthi…

CASE CONNECTION

BD just returned from a camping trip in the mountains. "I wish I was here to show you travel pictures, but I'm here about diarrhea. I've had nausea and cramps; my stomach is bloated. My stools smell terrible, and I've had a lot of gas. And I'm just plain tired." BD has just told you pretty much everything you would have asked. He asks you, "Can you give me something for this?" His abdomen is mildly tender and distended; there is no rebound or organomegaly.

What can you give BD to treat his condition? Consider your answer as you read, and we'll revisit BD at the end of the brick.

GO TO CONCLUSION

The mechanism of action for most antiparasitic and antihelminthic drugs is selective for cellular processes unique to the parasite. These drugs do not cause the same metabolic response in humans. As a result, the medications are well tolerated overall. This ability to exclusively act on the pathogen without having toxic effects to the host is known as selective toxicity.

What Are the Drugs Used to Treat Intestinal Protozoal Infections?

Protozoal disease is not just confined to the gut; a number of such illnesses infect other parts of t … rious is malaria, which infects the red blood cells (RBCs) and has caused millions of deaths worldwide. However, we will discuss antimalarials in detail in the Heme course. In this brick, we will discuss other protozoal infections.

How do some commonly used Antiprotozoal agent work? (LO#1 and 3)

Metronidazole

Metronidazole is the most commonly prescribed antiprotozoal agent. Metronidazole is absorbed by passive diffusion in its unionized form. The drug is active when it is reduced intracellularly by electron transport proteins such as ferrodoxin. Anaerobic bacteria have the electron transport necessary for reduction. Metronidazole is reduced by the anaerobic bacteria; the reduced metronidazole and the free radicals interfere with DNA synthesis and can result in DNA degradation. Anaerobic bacteria has the transport chain and hence provide coverage for anaerobic infections. These mechanisms give Metronidazole unique versatility in treating gram-negative bacterial infections. It also is effective against many protozoal infections. It is the drug of choice in several infections (e.g., trichomoniasis, amebiasis, and giardiasis).

Why is Metronidazole the drug of choice for trichomoniasis, amebiasis, and giardiasis?

Metronidazole is mostly absorbed in the intestinal tract and is partially excreted through bile. So, it acts not only in the tissues but also in the intestinal lumen.

The drug is generally well tolerated with the most **common adverse effects** being headache, nausea, and metallic taste. Alcohol should be avoided during and 3 days after use. It can cause a disulfiram-like reaction (Disulfiram-like reaction can cause flushing, tachycardia, nausea). It is contraindicated in the first trimester of pregnancy.

CLINICAL CORRELATION

Disulfiram was used in the past to treat alcohol dependence. It increases sensitivity to alcohol by inhibiting acetaldehyde dehydrogenase. As a result, acetaldehyde accumulates and gives patients the symptoms of alcohol intoxication (eg, nausea, vomiting, headache) as a deterrent to drink. Drugs with disulfiram-like reactions either inhibit acetaldehyde dehydrogenase or act in other ways to slow alcohol metabolism.

What should be avoided when taking Metronidazole?

Pentamidine

The mechanism of action involves disruption of microbial RNA/DNA, phospholipid, and protein synthesis by interference with nucleic acid incorporation into RNA and DNA. Adverse reactions includes reversible nephrotoxicity, pancreatitis, and acute hypotension.

Pyrimethamine

Pyrimethamine inhibits parasitic dihydrofolate reductase, resulting in inhibition of vital tetrahydrofolic acid synthesis. Patients might experience significant folate deficiency.

Which protozoal infections can these agents treat? (LO#2)

The Microbiology bricks on protozoal infections have discussed the presentation of these infections. Within endemic areas, no treatment is needed if the patient … areas, infection must be treated even i … Metronidazole can be used to treat infection that spreads past the GI tract to other parts of the body. It is the drug of choice for giardiasis and intestinal or extraintestinal trichomoniasis. Tinidazole has activity against amebiasis, giardiasis, and vaginal trichomoniasis (STD) and may be better tolerated than Metronidazole.

Pyrimethamine is commonly used to treat Toxoplasmosis. Trichomoniasis is another protozoal disease treated with a combination of Pentamidine and Tinidazole.

What is the drug of choice for treating Giardiasis?

Summary of the Antiprotozoal agents

Drug and its common indications	Action	ADRs
Metronidazole/Tinidaz… *Giardia, Trichomonas*	… synthesis, and cause cell death	…, Metallic taste, Disulfiram reaction with alcohol use
Pyrimethamine *Toxoplasma*	Blocks folate metabolism by inhibiting dihydrofolate reductase, inhibiting nucleic acid synthesis	Folate deficiency
Pentamidine *Trichomonas*	Incorporated into DNA/RNA and protein synthesis	Nephrotoxicity, Pancreatitis

How Do We Treat Helminth Infections?

The microbiology bricks on Helminth parasites have detailed the life-cycle and clinical presentation of these organisms. We know that depending on the species, exposure to the adult organism or its eggs causes infection. For instance, most roundworms start off in the gut, but helminths can reside in a variety of organs, such as muscle, eye, and brain. Hence, treatment(s) will depend on the species and where it resides.

What are the three broad classes of helminths?

What are commonly used Anti–helminth agents and what do they treat? (LOs#1–3)

Common drugs for helminth infection include ivermectin, benzimidazoles, and praziquantel.

Ivermectin

Ivermectin acts by binding and inactivating specific glutamate-gated chloride channels in helminth (but not human) nerve and muscle cells; chloride influx leads to hyperpolarization and paralysis of nerve and muscle, leading to worm death. Ivermectin is the drug of choice for strongyloidiasis and onchocerciasis (river blindness, roundworm). It is also effective against some intestinal parasites, including *Ascaris* (roundworm) and *Enterobius* (pinworm). One notable **adverse effect** is the Mazzotti reaction: fever, lymphadenitis, urticaria, arthralgia, edema, and hypotension, which can occur during onchocerciasis treatment.

Why are there no neuromuscular adverse effects from ivermectin?

Albendazole and **Mebendazole**.

Albendazole and Mebendazole belong to the **Benzimidazole drug class.** (the drugs in this class have the suffix -bendazole). They inhibit formation of cytoplasmic microtubules in cells of intestinal helminths and larvae, causing glycogen depletion, decreased glucose uptake, accumulation of waste metabolites, and decreased adenosine triphosphate (ATP) production, eventually leading to energy depletion, immobilization, and worm death (with final clearance of immobilized worms by peristalsis). This class of medications is used to treat a broad range of helminth infection, including neurocysticercosis (tapeworm), echinococcosis (tapeworm), ascariasis (roundworm), and necatoriasis (hookworm). **Side effects** of albendazole include abdominal pain, nausea, vomiting, and increased hepatic transaminases; these are generally transient and usually do not require discontinuation of the drug.

What is t … dazoles?

Mnemonic

Benzimi**d**azoles treat **bend**y worms (helminthic infections).

Praziquantel. The agent works by increasing permeability in the susceptible worms which can lead to worm paralysis. The drug of choice for schistosomiasis and intestinal tapeworms.

What Are Characteristics of Antihelminthic and Antiprotozoal Therapies?

Summary of Anthelminthic infections

Drug and its common indications	Action	ADRs
Ivermectin *Onchocerca, Strongyloides*	Increases intracellular chloride, causing worm muscle paralysis	Fever, lymphadenitis, urticaria, arthralgia, edema, hypotension while treating onchocerciasis
Praziquantel (*Schistosomiasis*)	Increases in permeability leading worm muscle paralysis	Itching, headache, dizziness
Benznidazole Intestinal worms (*Necatoriasis, Ascaris*)	Inhibits synthesis of DNA, RNA, and proteinss	

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to BD, what can you give him to treat his condition?

BD is a very helpful patient, laying out a fairly complete picture of his condition. You explain, "I think you are infected with giardia. It's a parasite you can catch by ingesting it. Did you drink any water from a mountain stream, perhaps one that the beavers liked to visit?" BD nods. "Beavers are a notorious carrier of these parasites. We can do some stool tests to prove the diagnosis, but I'm convinced enough that we could just go ahead and treat you." You and BD discuss starting treatment versus getting tests first and decide together to go ahead and treat him using metronidazole. In this kind of situation with a straightforward history, adding diagnostic tests delays treatment and adds cost but doesn't necessarily change what you do for the patient.

Summary

- Selective toxicity is the ability of a drug to act exclusively on the pathogen without having toxic effects in the host. This is the mechanism of action for most antiparasitic and antihelminthic drugs.
- Metronidazole (and Tinidazole) is the most commonly prescribed antiprotozoal agent. It is effective against most protozoal infections such as Trichomoniasis and Toxoplasmosis.
- Pentamidine and Pyrimethamine are also used in the treatment of protozoal infections such as Toxoplasmosis.
- Helminths are parasitic worms that may reside in a variety of organs. Treatments for helminth infections depend on the species and where it resides.
- Albendazole and Mebendazole disruption protein, DNA and RNA synthesis and cause worm death. They are commonly used in most helminthic infections and can cause hepatotoxicity.
- Ivermectin depletes metabolic reserves (glucose/glycogen) and causes worm death. They are used in treatment of roundworm and *Strongyloides* infections.

Review Questions

1. Metronidazole will not be effective against which one of the following infections?

A. ○ Amebiasis
B. ○ Giardiasis
C. ● Ascariasis
D. ○ Trichomoniasis

Hide Explanation

The correct answer is Ascariasis(C), which is a helminthic infection that cannot be treated with metronidazole. Antihelminthic agents such as Albendazole would be effective treatment for an Ascariasis (Roundworm) infection. Amebiasis (A), anaerobic bacteria (B), giardiasis (C), and trichomoniasis (E) can all be treated with metronidazole.

2. A patient who recently returned from international travel complains of episodic abdominal pain and a recurrent cough. Stool testing is positive for larvae suggestive of strongyloidiasis. Which of the following drugs is most useful for this patient?

A. ○ Tinidazole
B. ● Ivermectin
C. ○ Metronidazole
D. ○ Primaquine

Hide Explanation

The correct answer is ivermectin (B). Strongyloidiasis is contracted from walking on contaminated soil. The helminths spread to both intestines and lung. The preferred drug is ivermectin. Benznidazole (A) is the drug of choice for *T cruzi*, the pathogen responsible for Chagas disease. Metronidazole (C) is used primarily in protozoal infections. Primaquine (D) is an antimalarial that is only used in uncomplicated malaria to reduce transmission rates or when first-line therapy fails.

3. Which drug should be spaced from the use of alcohol?

A. ○ Albendazole
B. ○ Praziquantel
C. ○ Ivermectin
D. ● Metronidazole

Hide Explanation

The correct answer is metronidazole.(D). Metronidazole, when taken with alcohol can cause a disulfiram typ reaction and hence should be avvoided. Alcohol should be avoided with albendazole, Ivermectin have interactions with alcohol, but do not cause a disulfiram-type reaction.

4. Which agent acts as an anthelmintic by depleting metabolic function and microtubule disruption?

A. ○ Metronidazole
B. ○ Pyrimethamine
C. ○ Ivermectin
D. ● Albendazole

Hide Explanation

The correct answer is Albendazole (D). Albendazole can deplete glucose and glycogen reserves, causing worm starvation. It also disrupts microtubule formation, leading to worm paralysis. Metronidazole (A) is the incorrect answer since metronidazole is an antiprotozoal agent that increases free-radical formation and DNA damage. Pyrimethamine (B) is incorrect since it is an antiprotozoal agent. Ivermectin (D) disrupts chloride influx and causes worm immobilization.

How would you rate this Brick?

👍 👎

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How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action of antihelminthic and antiprotozoal drugs.~~
02 ☑ ~~List the indications for anthelminthic and antiprotozoal drugs.~~
03 ☑ ~~Recognize the common adverse reactions associated with anthelminthic and antiprotozoal drugs.~~

Objective Confidence: 3/3